

[Turn over

2

For A: $T = (3.0)a$ (1)

For B: $(7.0)(9.81) - T = (7.0)a$ (2)

Sub (1) into (2):

$$68.67 - 3a = 7a$$

$$a = 6.9 \text{ m s}^{-2}$$

Answer: B

5. When the mass is fully submerged in the water, the sum of the weight Mg acting on the